

Unit

3

Lesson

2



Beyond Imagination.

Speak Up, Smart Bin!

Time Frame: [90 min].

Ready, Set, Learn!

You Need to Know

- Explain how sound and light help people understand information.
- Use Arduino to send messages using LCD screens.
- Understand how solar panels help power smart devices.

challenge questions of the lesson.

1. Imagine your Smart Bin's LCD screen breaks, but it still needs to tell you important things, like when it's full. What are two other ways the bin could communicate messages to you using different outputs you've learned about (lights or sounds)?
2. If your Smart Bin is powered by a solar panel and a battery, what is the main benefit of using solar energy for a device like this? What might be a challenge for a solar-powered bin if it's placed in a very shaded area?

SEL Relation

1. Social Awareness (Empathy and Perspective-Taking):

- By designing a smart bin or other smart tech solutions for everyday problems, students engage in perspective-taking, imagining the inconveniences or challenges people face in daily life (e.g., waste management, energy consumption).
- They consider how their technological solutions can improve convenience and efficiency for users and contribute to environmental sustainability.

2. Responsible Decision-Making:

- Students engage in ethical consideration by designing solutions that address real-world problems and promote sustainability.
- They make responsible choices when planning their smart tech solution, considering user needs, environmental impact, and safety during the circuit assembly and testing process.
- Problem-solving skills are honed as they troubleshoot their circuits and adjust their designs, demonstrating persistence and adaptability.

3. Relationship Skills (Collaboration):

- Working in pairs or small groups during the design and building phases strengthens collaboration and communication skills, as they share ideas and work together to integrate components.

Engage



Teacher Role:

- Problem Identifier: Guide the discussion about the issue of the trash bin overflowing when its fullness isn't known.
- Idea Facilitator: Encourage and affirm students' brainstorming ideas, especially those related to using sensors (like IR) and visual alerts (LEDs, LCD) for notification.
- Questioner: Pose open-ended questions to deepen their thinking about practical solutions for detecting bin fullness and displaying that information clearly.
- Connector: Link their ideas (like using an IR sensor and LEDs) to the concept of wiring these components to the Arduino, setting the stage for the next hands-on activity.
- Presenter: Introduce or clarify the basic function of IR sensors and LEDs as components in a smart alert system.

Expected From Students:

- Active Listening: Students will listen attentively to the dialogue and identify the problem of the bin overflowing.
- Problem Identification: Students will understand why a full bin without an alert system is a problem for Mr. Yusef and the schoolyard.
- Idea Generation: Students will brainstorm initial ideas for how a sensor could detect fullness and how lights or screens could show this information.
- Participation in Discussion: Students will share their thoughts, suggest sensor and display solutions, and respond to teacher and peer questions.
- Critical Thinking (Initial): Students will begin to think about how different components (IR sensor, LED, LCD) contribute to a smart alert system.

Parts of the books included.



Ms. Sara: We've run into a problem—Mr. Yusef doesn't realize the bin is full until it starts overflowing. So, we're still dealing with the same old issue!

Adam: We need a way to warn him before it gets that bad.

Laila: Maybe we can use an IR sensor to tell when the trash reaches the top.

Ms. Sara: That's a great idea! But how can we make sure the message is clear?

Zayn: We could use LED lights—green means empty, red means full!

Adam: Or we can take it further and add an LCD screen that shows messages like "Bin Empty" or "Bin Full."

Ms. Sara: Excellent ideas, everyone! Let's begin by connecting the LEDs to our Arduino."

STEAM SCOUTS

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Let's Think!

1. Can you think of five different things where lights or screens are used to give people important information?

2. Imagine all the electricity in the world suddenly disappears! Our awesome smart bin can't use its sensors or motors anymore. How can we make a smart bin work all by itself, automatically, even without any electricity?

Story



1. Set the Scene with Imagination:

- Teacher says:
 - "Close your eyes for a moment and imagine you're Mr. Yusef, the person who helps keep our schoolyard clean. You're busy, and suddenly, a trash bin overflows without you knowing! How would that feel? How would you wish you knew it was full before it became a big mess?"

2. Engage with the Problem Dialogue:

- Purpose: This section uses the provided dialogue as a mini-play or shared reading to immerse students in the bin's "fullness" problem and introduce the concept of sensor-based alerts.
- Implementation:
 - Read Aloud with Roles: Assign roles (Mrs. Sara, Adam, Laila, Student 1, Student 2) and read the dialogue aloud with expression. Encourage different voices for each character (e.g., concerned for Mrs. Sara, enthusiastic for Adam, thoughtful for Laila).

3. Pause and Discuss: After key lines, pause and ask guiding questions:

- "What's the main problem Mrs. Sara, Adam, and Laila noticed about the full bin?"
- "What's Laila's idea to detect when the trash reaches the top?"
- "What exciting ideas do the students and Adam have for showing that information?"

4. Prompt a Thinking Question:

- Teacher asks: "What could we design to help Mr. Yusef know when the bin is full before it overflows? What kind of smart alert system, using sensors and lights, could we create?"
- Let students share quick ideas to activate prior knowledge about alerts or sensors.

Possible Strategies for Implementation:

1. "Mystery Component" Prediction & Discussion

- How to implement: Show students actual IR sensors and LEDs (or large, clear images if real components aren't available). Ask guiding questions: "Based on our discussion about the bin problem, how do you think this component [point to IR sensor] could help the bin 'see' if it's full?" and "How could this component [point to LED] show Mr. Yusef that information?" Encourage students to make predictions before confirming the basic functions of each component.
- Purpose: Directly engages students with the physical (or visual) components mentioned in the dialogue, making abstract concepts more concrete and fostering

curiosity about how these parts work in a smart system.

2. "Draw Your Smart Alert Idea"

- How to implement: Provide students with a simple worksheet or drawing paper. Ask them to draw their version of the "smart alert" system for the bin. Encourage them to focus on how the IR sensor might detect the trash level and how the LEDs (or even a simple drawing of an LCD screen) would clearly signal "full" or "empty." They can label the different parts of their drawing.
- Purpose: Allows students to visually process and communicate their understanding of the proposed solutions, reinforcing their "Idea Generation" and "Problem-Solving" skills in a creative and personal way.

3. "Bin Status" Role-Play

- How to implement: Designate an area as the "bin." One student can be the "IR sensor" (using their hand to "detect" if the bin is full by checking for an imaginary trash level). Another student can be the "LED" (holding up a green card for "empty" or a red card for "full"). Have a "Mr. Yusef" character respond to the signals. Change the "bin status" and have students demonstrate the detection and alert.
- Purpose: Provides a kinesthetic and highly engaging way for students to understand the flow of information from sensor to output in a smart system, enhancing "Problem Identification" and "Critical Thinking."

Let's Think!



Objective:

Students will be able to identify various real-world applications of lights and screens for conveying information, and creatively brainstorm non-electrical solutions for automated systems.

Answers:

Examples of Lights and Screens for Information.

- Traffic Lights: Use colors (red, yellow, green) to tell drivers and pedestrians when to stop or go, helping them move safely.
- Smartphones/Tablets/Computers: Their screens display messages, pictures, videos, and apps that help us learn, communicate, and navigate.
- ATMs (Automated Teller Machines): Screens show bank account information and guide us through transactions like withdrawing money.
- Store "Open/Closed" Signs: Simple lights or screens tell us if a business is ready for customers.
- Alarm Clocks/Digital Watches: Screens display the time, helping us stay on schedule.



Let's Think!



1. Can you think of five different things where lights or screens are used to give people important information?

2. Imagine all the electricity in the world suddenly disappears! Our awesome smart bin can't use its sensors or motors anymore. How can we make a smart bin work all by itself, automatically, even without any electricity?

- Washing Machines/Microwaves/Ovens: Lights and screens show cycles, time remaining, or temperature.
- School Smartboards/Projectors: Project screens display lessons, videos, and interactive content for learning.
- Elevator Floor Displays: Lights or screens show which floor the elevator is on or which way it's moving.

Non-Electrical Smart Bin Solutions.

If all traditional grid electricity disappeared, for our smart bin to still work automatically, we would need to find ways for it to generate its own power or rely on purely mechanical actions:

- Solar Power: We could attach a solar panel to the bin. The sun's energy would be captured by the panel and converted into electricity to power the bin's sensors and motors, allowing it to work automatically without needing to be plugged into a wall outlet.
- Wind Power: Similar to how large wind turbines generate electricity, a small turbine could potentially be built into or near the bin to generate power from the wind, making it self-sufficient.

Alternatively, if absolutely no electricity of any kind were available, we would need to think of purely mechanical solutions like:

- Weight-Activated Lid: A lid designed with a counterbalance or spring that opens automatically when a certain weight of trash pushes on a lever inside.
- Foot Pedal Mechanism: A pedal that, when pressed, uses a system of levers and hinges to lift the lid without any power source.
- Gravity-Assisted Sorting: Different compartments could have angled ramps or funnels that sort trash based on its weight or how it slides, directing items into different bins below using only gravity.

Possible Strategies for Implementation:

1. "Information Everywhere!" Brainstorm & Share (for Question 1).

- How to Implement: Begin by having students close their eyes for 30 seconds and recall all the places they've seen lights or screens give them information in their daily lives (e.g., at home, school, in cars, stores). Then, have them open their eyes and quickly list as many as they can for Question 1. Afterward, facilitate a class share-out, adding unique ideas to a collaborative list on the board.
- Purpose: Activates prior knowledge about ubiquitous visual cues and broadens awareness of how information is conveyed in the real world.

2. "No-Power Automation" Design Challenge (for Question 2)

- How to Implement: Introduce the hypothetical scenario dramatically. Divide students into small groups or have them work individually. Challenge them to sketch or describe a smart bin system that operates automatically without any electricity. Encourage them to think about simple machines, gravity, levers, weights, or other mechanical triggers that could make a lid open or sort trash.

- Purpose: Fosters creative problem-solving and critical thinking by requiring students to innovate under constraints, reinforcing the concept of automation while moving beyond electrical reliance.

Explorer's Toolkit



Teacher Role:

- **Introducer & Vision Setter:** Clearly explain the purpose of the activity: to bring their code messages to life on a digital screen, just like many real-world devices.
- **Tinkercad Navigator & Circuit Guide:** Provide clear, step-by-step instructions for navigating Tinkercad, finding the LCD and Arduino components, and making the correct virtual connections.
- **Code Uploader & Syntax Helper:** Guide students on where to find and paste the provided code, and help them understand basic syntax or common errors if they type.
- **Observer & Debugging Coach:** Circulate and observe student progress. Help students troubleshoot (or "debug") their virtual circuits and code if they encounter errors, guiding them to find solutions.
- **Discussion Facilitator:** Lead a "Let's Talk About It" discussion, prompting students to reflect on their experience with the LCD and what it means to see code in action.
- **Logical Thinking Encourager:** Prompt students to think about how code creates specific messages and why clear instructions are important for the LCD to display correctly.

Expected From Students:

- **Active Listening:** Students will listen carefully to instructions for building the circuit and working with code in Tinkercad.
- **Careful Virtual Building:** Students will follow accuracy rules when connecting components in Tinkercad to build their LCD circuit.
- **Code Completion & Debugging:** Students will utilize tools (AI tool) to identify and complete missing parts of the code, and troubleshoot to ensure correct functionality.
- **Code Application:** Students will accurately paste/upload code and run simulations to see their messages.
- **Observation & Analysis:** Students will observe what messages appear on the LCD and analyze how changes in code might affect the display.
- **Reflection & Articulation:** Students will reflect on their experience and answer questions about how they made the LCD show their code's messages.

Parts of the books include.

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Set the LCD address to 0x27 for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.begin();           // Initialize the LCD
  lcd.backlight();       // Turn on the backlight

  lcd.setCursor(0, 0);   // Set cursor to column 0, row 0
  lcd.print("Smart Bin"); // Print first line

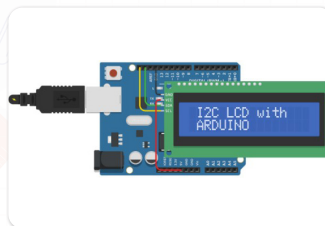
  lcd.setCursor(0, 1);   // Set cursor to column 0, row 1
  lcd.print("Ready to Work!"); // Print second line
}

void loop() {
  // Nothing needed here for now
}
```

DIY Lab

Get ready to build and test your LCD circuit in Tinkercad—your message will soon show up on the screen!

Use an AI tool to search for the missing parts of the code, complete it, and then run your test.



```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Set the LCD address to 0x27 for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.begin();           // Initialize the LCD
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  lcd.setCursor(0, 0);   // Set cursor to column 0, row 0
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  lcd.setCursor(0, 1);   // Set cursor to column 0, row 1
  lcd.print("Ready to Work!"); // Print second line
}

void loop() {
  // Nothing needed here for now
}
```

Figure 32: LCD Display Circuit with Arduino.

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DIY Lab

Objective:

Students will explore how code messages can be displayed visually on an LCD screen in Tinkercad by completing and debugging pre-prepared code snippets. They will learn to build a virtual circuit, work with code, and observe its output, fostering an understanding of digital communication and troubleshooting.

Materials Needed:

- Laptops.
- Internet Access: Required for Tinkercad.
- Tinkercad Accounts: Pre-created or guidance for students to sign up.
- Pre-prepared LCD Code Snippet: A simple code to display "Hello World!" or "Smart Bin!" on the LCD.
- (Optional) Projector or large screen for whole-class demonstration of Tinkercad steps.

Activity Instructions:

- Set Up the Tinkercad Workspace:
 - Guide students to open Tinkercad Circuits.
 - Show them how to create a new circuit design.
 - Help them locate the Arduino Uno, and the LCD (specifically, the 16x2 LCD module) in the component's library.
- Assemble the Virtual Circuit:
 - Provide clear, step-by-step instructions for connecting the LCD to the Arduino
 - Emphasize correct connections for power (5V, GND) and data pins.
 - Connections:
 1. SDA (Serial Data Line) → A4 on Arduino
 2. SCL (Serial Clock Line) → A5 on Arduino
 3. VCC → 5V on Arduino

4. GND → GND on Arduino

- Prepare the Code Challenge: Provide students with a version of the LCD code snippet that has specific lines or sections intentionally removed, requiring them to use tools or knowledge to complete it. (The full code is provided for your reference below).
- Code (Full Snippet for Reference):

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Set the LCD address to 0x27 for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.begin();      // Initialize the LCD
  lcd.backlight();  // Turn on the backlight
  lcd.setCursor(0, 0); // Set cursor to column 0, row 0
  lcd.print("Smart Bin"); // Print first line
  lcd.setCursor(0, 1); // Set cursor to column 0, row 1
  lcd.print("Ready to Work!"); // Print second line
}

void loop() {
  // Nothing needed here for now
}
```

- Upload the Code:
 - Explain where to find the "Code" editor in Tinkercad.
 - Show them how to start the simulation.
- Test and Observe:
 - "What message do you see? Does it match what we put in the code?"
 - Challenge them to change the message in the code and re-run the simulation.
- Reflection Questions (Let's Talk About It):
 - How did the LCD help you "see" what the code was doing?
 - What was the most important step in making the message appear on the screen?
 - Why do you think smart devices, like our smart bin, sometimes have screens to show messages?

Possible Strategies for Implementation:

- **Model First:** Demonstrate each step of building the circuit and uploading the code on a projected screen before students try it themselves.
- **Step-by-Step Instructions:** Provide clear, visual step-by-step instructions (on a handout or projected slide) for building the circuit in Tinkercad.
- **Common Errors Chart:** Prepare a small chart of common errors ("wires not connected," "spelling mistake in code") and how to fix them.
- **Debugging Challenge:** After they get "Hello World!" working, challenge them to change the message or display two different messages.
- **Connect to Real-World:** Show examples of real-world devices that use small screens to display information (e.g., microwave).

Explain



Teacher Role:

- **Introducer & Demonstrator:** Introduce the fully integrated Smart Bin circuit (built by the teacher) and demonstrate its functions. Connect this physical demonstration to the concept of automated actions.
- **LCD Expert & Circuit Guide:** Provide clear, step-by-step instructions for navigating Tinkercad, building the LCD circuit, uploading code, and interpreting the displayed messages.
- **Solar Panel Explainer:** Facilitate a discussion on how solar panels generate electricity, linking it to the smart bin's potential for sustainable operation.
- **Clarifier & Troubleshooting Coach:** Explain core concepts (like sensor input, output, and how components interact). Help students troubleshoot (or "debug") their virtual LCD circuits if they encounter errors.

Questioner: Prompt students to explain their thinking with questions like:

1. "What did the smart bin do when I put my hand here?"
 2. "How does the bin 'know' if the trash is wet or dry?"
 3. "Why is it important for the bin to sort trash automatically?"
 4. "How does the LCD help us 'see' information from the bin?"
 5. "How could we use solar panels to power our smart bin?"
- **Connector:** Help students connect their observations from the Smart Bin demonstration and their experiences with the LCD activity back to real-world smart devices, the "Turning Trash into Power" concept, and sustainable energy solutions.

Expected From Students:

- **Attentive Observation:** Students will carefully observe the teacher's demonstration of the integrated Smart Bin circuit's functionality.
- **Conceptual Understanding of Automation:** Students will begin to understand how sensor inputs lead to automated outputs in the smart bin.
- **Identification of Inputs & Outputs:** Students will identify how sensor inputs (e.g., trash detected, fullness, moisture) lead to specific outputs (LED on, lid open, sorting).
- **LCD Circuit Building & Code Application:** Students will build a virtual LCD circuit in Tinkercad and apply provided code to display messages on the screen.
- **Observation & Analysis of LCD Output:** Students will observe what messages appear on the LCD and analyze how the displayed information relates to the circuit.
- **Understanding of Solar Panels:** Students will begin to grasp the basic principles of how solar panels generate electricity and their potential applications for smart devices.
- **Active Communication:** Students will share their observations from the smart bin demonstration, their experiences with the LCD activity, and their ideas about solar energy integration during discussions.

Parts of the books include.



Screen Lessons

Let's learn more about how solar panels generate electricity.

https://www.youtube.com/watch?v=ZzCjZb8mFwM&ab_channel=Which%3F



Screen Lessons

Objectives:

Students will learn about how solar panels can be used to power devices like the smart bin. This activity encourages critical thinking, reinforces understanding of digital communication, and introduces the concept of sustainable energy.

Video link: https://www.youtube.com/watch?v=ZzCjZb8mFwM&ab_channel=Which%3F



Possible Strategies for implementation:

1. Pre-Watching: Set a purpose.

- **Strategy: "Energy Mission":** Before playing the solar panel video, tell students: "Today, we're going on an 'energy mission' to make our smart bin even better! The video will help us remember or learn about solar panels, which are super important for

generating electricity and powering devices." Ask: "What do you already know about energy from the sun? What about how things get power?"

- Benefit: Frames the video within the context of their STEAM project and activates prior knowledge about energy.

2. During Watching: Active engagement.

- Strategy: "Trace the Power/Identify": Encourage students to trace along or physically mimic the "generating" (making power) and "transferring" (sending power) actions shown in the video. If the video uses visuals, they can trace how the electricity flows.
- Benefit: Promotes active engagement with the concepts presented in the video.

3. Post-Watching: Discussion and reflection.

- Strategy 1: "Recap the Solar Power": Ask questions like: "What did the video teach us about 'solar panels'? What about 'generating electricity'?" Use their observations to confirm their understanding: "Remember, solar panels mean putting sunlight together to make power."
- Benefit: Consolidates the learning from the video.

Elaborate



Teacher Role:

- Facilitator: Introduce the questions as a way to review and reinforce understanding of LCDs and solar panels. Then, introduce the "Build and Test Your Brilliant Solar Panel with an LED!" activity as a practical way to see solar energy in action.
- Guide: Support students as they transition from answering questions to the hands-on activity. Help them understand how their knowledge of solar panels applies to building a simple circuit that converts light into electricity.
- Prompt and Support: Offer helpful questions and encouragement as students connect the solar panel, switch, and LED. Guide them to observe how the LED lights up with solar power and to troubleshoot if the circuit doesn't work as expected.
- Observer & Assessor: Watch for how students engage with the questions and apply their understanding in the solar panel circuit building activity. Note their accuracy in answering questions, proper circuit connections, and observations of the LED lighting up.
- Motivator: Celebrate students' successful circuit building, their understanding of clean energy, and highlight the importance of hands-on application and sustainable thinking.

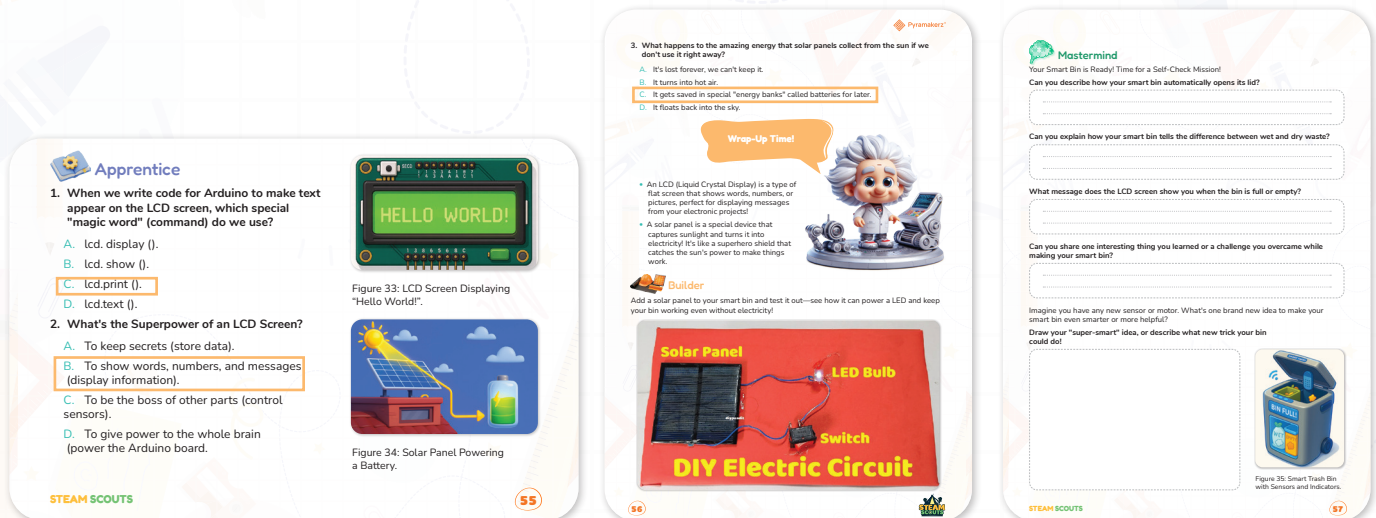
Expected From Students:

- Demonstrate Understanding: Students will answer questions to show what they have learned about LCD screens and how solar panels generate energy.
- Apply Knowledge: Students will use what they learned by building and testing a

simple electric circuit using a solar panel and an LED.

- **Follow Instructions & Connect:** Students will carefully follow directions to connect the solar panel, switch, and LED, ensuring the circuit is correctly assembled.
- **Observe & Test:** Students will actively test their circuit, observe the LED lighting up, and understand that solar energy is making it work.
- **Share & Explain:** Students will be able to show their working solar panel circuit and explain how the solar panel provides power to the LED.

Parts of the books included

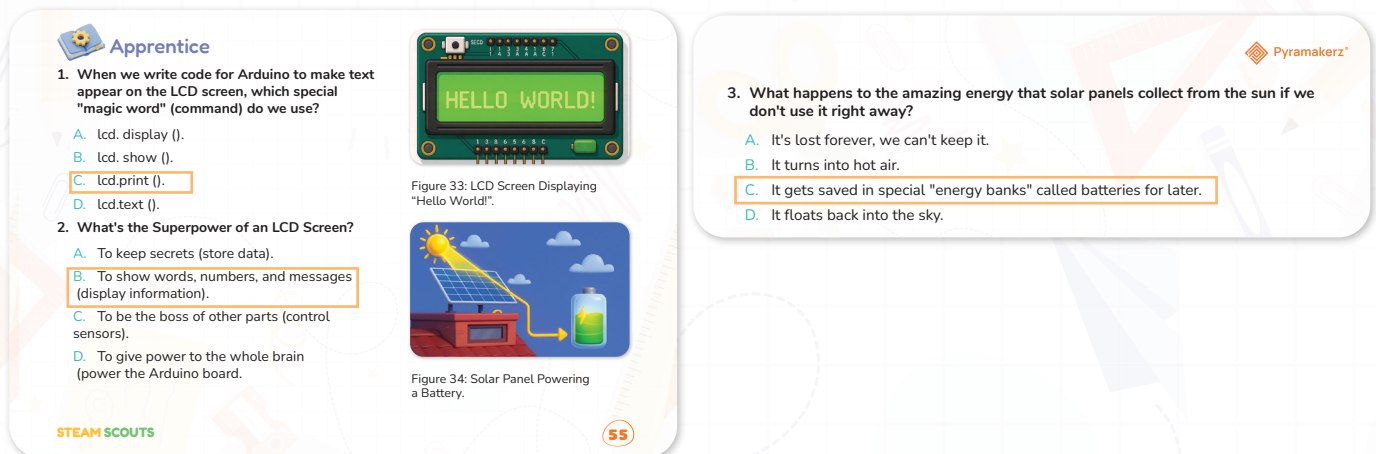


Apprentice

Objective:

Students will review and reinforce their understanding of how LCD screens display information and how solar panels generate electricity by answering targeted questions. This activity strengthens conceptual understanding, recall, and critical thinking skills needed when programming visual outputs or designing sustainable power solutions for smart devices.

Answers:



Possible Strategies for Implementation:

1. Set a Purpose Before Answering Questions.

- Tell students: "These questions will help us quickly review and check our understanding of how LCDs show messages and how solar panels make electricity. You'll need this knowledge as we continue to understand and improve our Smart Bin!"
- You can write on the board: "Today I'm reviewing smart tech to become a better innovator!"

2. Use as a Warm-Up or Knowledge Check.

- Let students discuss or answer these questions in pairs or independently for 5–10 minutes to boost recall before moving into a related building or discussion task about the Smart Bin.

3. Timed Challenge (Optional).

- Challenge students to answer the questions in a set time, then review their answers as a class. This builds quick recall and focus.

4. Smart Tech Review Prompt.

- After answering the questions, ask students to answer in a quick journal or share: "Which question about LCDs or solar panels was most interesting? Which one was the trickiest for you to answer, and why?" This builds metacognition.

5. Connect to Real-World Task.

- After reviewing, transition to a hands-on activity or deeper discussion by saying: "Now that you've refreshed your smart tech knowledge, let's use it to think more about our Smart Bin! You'll need to remember how LCDs work to understand messages and how solar panels provide power to make our bin sustainable."



Builder

Objective:

Students will build and test a simple electric circuit using a solar panel, a switch, and an LED, to understand how solar energy is converted into electricity to power a light source. This activity helps students apply basic circuit building principles, observe energy conversion, and fosters problem-solving and hands-on skills in renewable energy.

Possible Observations/Answers:

- "The LED lights up when the solar panel is in bright sunlight and the switch is on."
- "The LED doesn't light up if it's dark, even with the switch on."
- "The solar panel changes light energy into electricity."
- "The switch turns the light on and off, controlling the electricity flow."
- "My circuit needs sunlight to work."
- Activity Reference: https://www.youtube.com/watch?v=uzZf_I2SsMQ



Activity Instructions:

1. Introduce the Goal.

- Say: "Today, we're going to build a simple circuit that uses the sun's power to light up an LED! We'll see how solar panels can make electricity, just like we learned in the video."

Materials Needed.

- Solar panel
- LED bulb
- On/off switch.
- Connecting wires
- A base material (cardboard, foam board).
- Strong light source (flashlight, sunlight from a window).

Guide Step-by-Step.

- Prepare the Base: Have students secure their solar panel, switch, and LED bulb onto the cardboard base.
- Make Connections: Guide students to connect the components with wires, following a simple circuit diagram:
 - Connect one wire from the solar panel's positive (+) terminal to one terminal of the switch.
 - Connect another wire from the other terminal of the switch to the longer leg (positive) of the LED.
 - Connect a final wire from the shorter leg (negative) of the LED back to the solar panel's negative (-) terminal.
 - Emphasize LED Polarity: Remind students that LEDs only work if connected the correct way (long leg to positive, short leg to negative).
- Test the Circuit: Have students turn on the switch and expose the solar panel to a strong light source (like sunlight from a window or a flashlight).
- Observe: Ask them to observe what happens to the LED.

Possible Strategies for Implementation

1. Guided Discovery Connection.

- Instead of immediately showing the full circuit diagram, provide students with the components and challenge them, in pairs, to try and make the LED light up using the solar panel and switch. Offer hints or simplified sub-diagrams only if they get stuck, encouraging experimentation.

2. Predict & Test Cycle.

- Before building, have students draw what they think the correct circuit connections should look like and predict what will happen when they expose the solar panel to light. After building, they compare their predictions to their observations.

3. Troubleshooting Challenge.

- Prepare a few pre-assembled circuits with a common, easy-to-find mistake (e.g., reversed LED, a loose wire). Present these to students and challenge them to find and fix the "bug" to make the LED light up.

4. Solar Power Showcase

- Once circuits are working, have each pair demonstrate their working circuit to another pair or to the class. Encourage them to explain how the solar panel makes the LED light up and what role each part plays.



Mastermind

Objective:

Students will apply their understanding of the Smart Bin's functionality and components by answering self-reflection questions and brainstorming new design ideas. This activity will help them strengthen their ability to explain complex systems, reflect on their learning journey, and engage in creative design thinking for future innovations.

Activity Instructions:

- 1. Introduce the Mission:** "You've worked incredibly hard on your Smart Bin project, learning about its 'brains,' 'muscles,' and 'eyes'! Now, it's time for a 'Self-Check Mission' to see everything you've learned and to imagine how we can make our bins even smarter!"
- 2. Clarify the Goal:** "Your mission is to answer these questions about your Smart Bin. Think about how it works, what special parts it uses, and what messages it shows you. Then, you'll get to be an inventor and dream up a brand-new idea!"
 - Can you describe how your smart bin automatically opens its lid?
 - Can you explain how your smart bin tells the difference between wet and dry waste? (Hint: What special sensor does it use for this?)
 - What message does the LCD screen show you when the bin is full or empty?
 - Can you share one interesting thing you learned or a challenge you overcame while making your smart bin?
- 3. Emphasize Reflective & Creative Thinking:** "This isn't a test! It's about remembering your journey and letting your ideas sparkle. Think about the 'how' and the 'why' of your bin's functions, and then let your imagination soar for new inventions!"
- 4. Introduce the "Super-Smart" Design Challenge:** "Now for the exciting part! Imagine you have any new sensor or motor you wish. What's one brand new idea to make your smart bin even smarter or more sensor or motor you wish. What's one brand new idea to make your smart bin even smarter or more."
 - Draw your "super-smart" idea, or describe what new trick your bin could do!

Possible Strategies for Implementation:

- 1. "Smart Bin Detective":** Refer to Your Work!

- How to Implement: Encourage students to look back at their previous notes, circuit diagrams, or even recall the demonstrations of the Smart Bin. Prompt them: "Think like a detective! Where can you find the clues to answer these questions?"
- Why it Helps: This strategy helps students access prior knowledge and reinforces the connection between the physical build, the code, and their understanding.

2. "Think-Pair-Share" for Reflection.

- How to Implement: Have students first think about their answers to the self-check questions individually. Then, pair them up to discuss their responses with a partner before sharing with the whole class (optional).
- Why it Helps: Promotes individual reflection, peer-to-peer learning, and helps students articulate their thoughts before a larger group.

3. "Idea Spark" Brainstorming for Innovation.

- How to Implement: Before starting the "super-smart" idea challenge, briefly brainstorm as a class:
 - "What are some other things sensors can do? What are some other ways motors can move things?" Give a quick example: "Maybe a sensor that knows if someone is close to the bin, not just right in front!"
- Why it Helps: Provides a scaffold for creative thinking, helping students generate ideas for their new smart bin features.

4. "Sketch Your Superpower": Visualize the Idea.

- How to Implement: Provide paper and drawing tools. Encourage students to draw their "super-smart" idea for the bin, emphasizing that simple sketches are great. They can label the new sensor/motor and what it does.
- Why it Helps: Allows students to express abstract ideas concretely, engages visual learners, and can simplify explaining complex concepts.

5. "Innovator's Pitch": Share Your Vision

- How to Implement: Have students briefly share their "super-smart" idea with a partner or the class. Encourage them to explain: "What's your new idea? What new sensor/motor does it use? How does it make the bin smarter or more helpful?"
- Why it Helps: Develops communication skills, encourages public speaking (even to a small group), and builds confidence in their innovative thinking.

Evaluate



Teacher Role:

- System Integrator Guide: Guide students through understanding how solar panels and batteries can be added to the Smart Bin system to make it self-sufficient and more independent. Use the provided images as reference for components like the battery charging module and general circuit connections.

- Sustainability Explainer: Emphasize and monitor the importance of designing systems that are "clean and green," highlighting how solar power contributes to a sustainable future.
- Problem-Solving Coach (Conceptual): Facilitate discussions around challenges in implementing self powering systems (e.g., storing energy, ensuring consistent power) and encourage students to brainstorm solutions.
- Encourager of Innovation: Stress the importance of thinking about energy efficiency and independence for smart devices.
- Evaluator: Assess students' understanding of how solar panels and batteries contribute to the Smart Bin's independence, as well as their ability to articulate the "clean and green" benefits.
- Discussion Leader: Prompt students to reflect on the advantages of a self-powered system, the role of batteries in energy storage, and how such designs contribute to a sustainable future.

Expected From Students:

- Conceptual Understanding: Students will describe how adding solar panels allows the Smart Bin to generate its own electricity.
- Energy Storage Comprehension: Students will explain the role of batteries in saving the collected solar energy for later use.
- System Integration Insight: Students will articulate how solar panels and batteries make the Smart Bin "smarter and more independent" by enabling self-power.
- Sustainability Awareness: Students will identify how a solar-powered Smart Bin contributes to a "clean and green future".
- Discussion Participation: Students will actively engage in discussions about designing for energy independence and sustainability.

Parts of the books included.



Showcase

To help the bin work even when it's dark, we'll connect the solar panel to a charge module, which safely charges a battery during the day. This stored energy powers the bin at night, making it self-sufficient, smarter, and better for a cleaner, greener future.

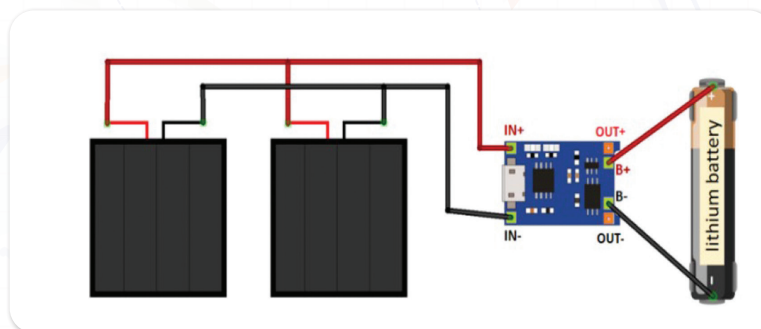


Figure 36: Solar-Powered Smart Bin Circuit.



Showcase

Objective:

Students will apply their understanding of smart system components and connections by physically assembling and integrating the main parts of their Smart Bin prototype. This activity reinforces knowledge of circuits and sensors, promotes hands-on problem-solving, and connects their design ideas to a functional, real world smart device.

- 1 x Solar Panel (6V) or 3 V.
- 1 x Rechargeable Battery Pack (Lithium or Lead-Acid).
- 1 x Charger Module.
- The whole circuit.
- Jumper Wires
- Multimeter (Optional for measuring voltage output).

Activity Instructions

Step 1: Connect the Solar Panel to the Charge Module.

- Identify the positive (+) and negative (-) terminals of the solar panel.
- Connect them to the input terminals on the charger Module.

Step 2: Connect the Battery to the Charge Module.

- Attach the battery pack's terminals to the output section of the charge controller.
- Ensure correct polarity (+ to +, - to -).

Step 3: Connect the Arduino System.

- Use the regulated output from the charge Module to power the Arduino.

Step 4: Test Under Different Light Conditions.

- Place the solar panel under bright sunlight and observe the LED.
- Cover the panel partially—does the LED dim?
- Test in low light—is the battery supplying power?

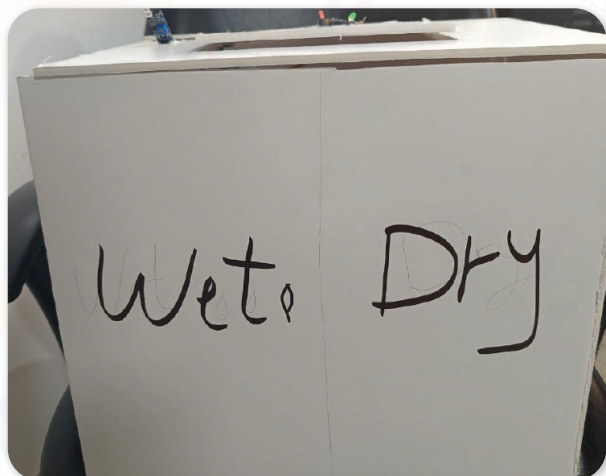
Step 5: Observe & Record Data.

- Use a multimeter to check voltage output from the solar panel and battery.
- Compare energy storage and usage at different times of the day.

Step 6:

- Once the main components are assembled and secured, have students power up their prototype and test its functionality.
- Encourage them to demonstrate how their bin automatically opens its lid, how it might simulate sorting, or what messages appear on their LCD screen.
- Prompt them to reflect on the process, any challenges they faced during assembly, and how their physical prototype fulfills its intended "smart" purpose.

Final Prototype:



Prototype video link: <https://drive.google.com/drive/folders/159MiNFZxF7xhcpjji-ugBcAUNikLPNDG>

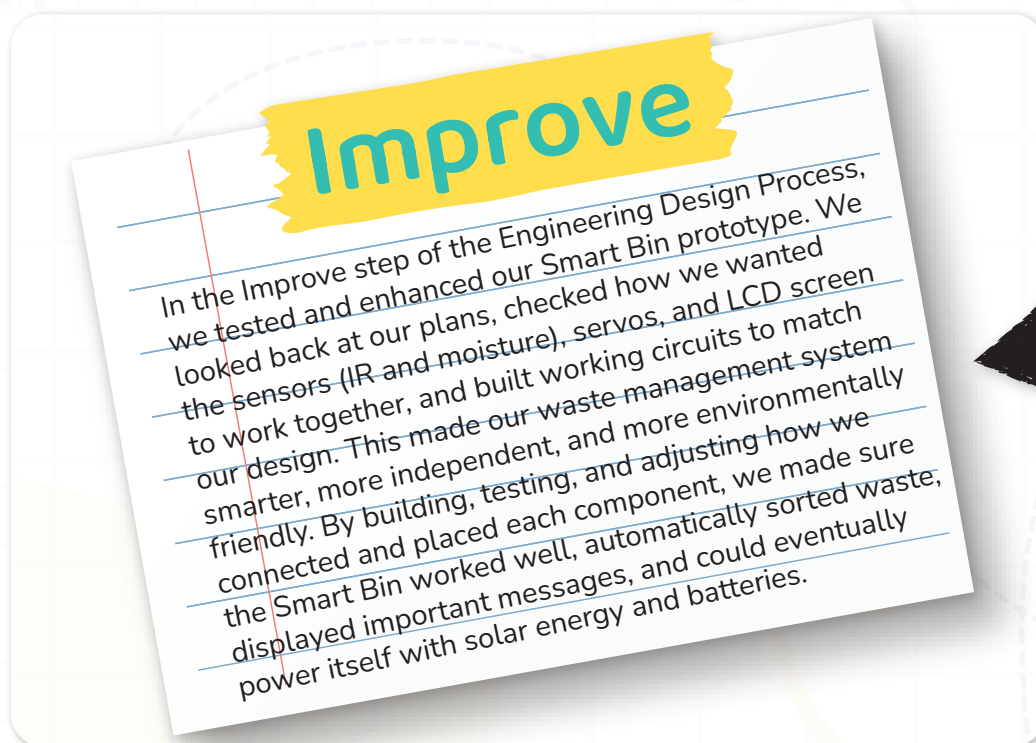


Rubric for Showcase Evaluation:

Criteria	Excellent (5 Point)	Good (3-4 Points)	Needs Improvement (1 Point)
Solar Panel Integration	Solar panel is correctly connected and powers the system efficiently.	Solar panel works but may need minor adjustments.	Solar panel does not power the system properly.
Battery Storage Functionality	The battery stores energy and provides power when needed.	Battery stores energy but has inconsistent performance.	Battery does not store power or is incorrectly connected.
LCD & LED Indicators.	The LCD displays correct status messages and LEDs work as indicators.	LCD and LEDs work but are missing some functions.	LCD and LEDs do not display or indicate correctly.
Sensor Integration	IR and moisture sensors work accurately for sorting and fullness detection.	Sensors work but may have small errors.	Sensors do not detect waste or bin fullness correctly.

Project Presentation	Student clearly explains how the system works and answers all questions confidently.	Student explains the system but struggles with some questions.	Student is unable to fully explain the system.
Problem-Solving & Improvements.	Student identifies challenges and suggests thoughtful improvements.	Student identifies some challenges but offers limited solutions.	Student does not discuss problems or improvements.

Don't forget to go back to the EDP page and add these to create part.



Now I Can

Objective:

Students will reflect on how they applied their understanding of smart technology, visual feedback (LCDs), and sustainable power (solar energy) in the Smart Bin project.

Activity Instructions:

1. Introduce the Purpose.

- Say: "Now that we have explored and worked on our Smart Bin project, let's take a moment to think about everything we learned and practiced."

- Explain that this activity helps students recognize what skills and knowledge they used in building and understanding the Smart Bin.

2. Read Each Statement Aloud.

- Read each (Now I Can) statement slowly and clearly.
- Pause after each one to give students a moment to think.

3. Student Response Options.

- Have students check the box next to each statement they feel confident about

4. Optional Sharing

- Let students share one (Now I Can) statement with a partner or in a small group.
- You may ask: "Which part of the Smart Bin (like the sensors, LCD, or solar power idea) do you feel you understand best now?" or "What was a challenge with the LCD or solar power that you overcame?"

5. Wrap-Up Reflection.

- Ask students to look at their checklist and reflect:
 - "Which skill do you want to practice more with smart devices?"
 - "How do you feel about building a smart device that can help with environmental sustainability?"

6. Collect and Review.

- Use the checklists to informally assess student understanding and guide review or celebration activities.

Relation to Standards:

This activity relates to various educational standards, including:

- Science/Physical Science (Electricity & Circuits): Understanding basic circuits, how components like sensors, LEDs, servo motors, LCDs, solar panels, and batteries interact within a circuit. Students learn that electricity flows in a closed loop.
- Engineering Design: Students engage in the "Build" phase of the engineering design process, turning their plan into a physical prototype. They also engage in testing and refining their design, integrating multiple components for automated functionality.
- Technology & Innovation: Using sensors, microcontrollers (Arduino), motors, and display screens to create a functional, automated, and interactive device. This fosters innovative solutions for waste management and sustainable energy.
- Social-Emotional Learning (SEL) - Social Awareness & Responsible Decision-Making: Fostering environmental responsibility by designing for efficient waste management and sustainable energy use. Students consider how their creation can make a positive impact on environmental sustainability. Students also practice responsible decision-making regarding safety and component care.



Mission Innovate

Final Mission – Innovate Your Daily Life with Smart Tech!

Overview:

This is a culminating, open-ended project that allows students to apply all the skills they've gained throughout the unit. Students will take full ownership of the process—from identifying a real-world everyday problem to designing a smart tech solution and presenting their idea.

Project duration: 2 weeks

- Teacher's Role:

- Facilitator of Inquiry:

Introduce the challenge by reading the mission prompt aloud. Help students understand that their task is to identify an everyday problem or inconvenience and design a smart tech solution to solve it, emphasizing convenience and efficiency.

- Encourager of Observation and Creativity:

Prompt students to think about the everyday problems or hassles they or others might face. Encourage them to brainstorm thoughtful smart technology solutions using sensors, automatic actions, and communication methods.

- Supporter of Student-Led Work:

Step back and let students take charge. Offer support only when needed—remind them to apply what they've learned about sensors, motors, coding logic, creating automated actions, and displaying information (like with LCDs or LEDs).

- Guide for Reflection:

Encourage students to use the reflection questions on their design worksheet and refer to the rubric during the design process.

- Assessment Guide:

Use the rubric provided to assess each group's final design/model based on science, technology, engineering, art (design), math (if applicable to their solution), and communication.

Implementation Tips:

- Let students work in pairs or small groups to foster collaboration.
- Have all previous materials and tools (e.g., craft supplies, reused items, Arduino, sensors, motors, LCDs, LEDs, wires, cardboard) available.
- Use “studio time” for students to design and build at their own pace.
- Allocate enough time across multiple sessions for ideation, design, potential prototyping, and presentation.
- Celebrate creativity and progress more than perfection.